

Big Data Analysis of Competitive Multiplayer Ranking Systems

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1 Introduction

Online competitive environments generate large volumes of data describing player activity, performance, and progression. The increasing availability of these data provides an opportunity to apply Big Data and Data Mining techniques in order to better understand competitive ecosystems and ranking dynamics.

This project focuses on the analysis of two independent competitive multiplayer communities that employ rating-based matchmaking systems. The study investigates how player performance relates to ranking progression and whether meaningful behavioral patterns can be identified within the player population.

The analysis follows the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology, covering data understanding, preparation, modeling, evaluation, and interpretation of the obtained results.

2 CRISP-DM Methodology

The project follows the CRISP-DM framework, a widely adopted methodology for structuring Data Mining projects.

The workflow is divided into five phases:

- **Business Understanding**
- **Data Understanding**
- **Data Preparation**
- **Modeling**
- **Evaluation**

The following sections describe how each phase is applied within the context of this study.

2.1 Business Understanding

The primary objective of this project is to investigate the behavior of competitive ranking systems using real-world player data.

In particular, the analysis aims to answer the following research questions:

- Which performance indicators are most strongly associated with player rating?
- Are there measurable characteristics that distinguish highly ranked players from the rest of the population?
- Can groups of players with similar behavioral profiles be identified?
- How do player ratings evolve over time?
- Are similar patterns observable across different competitive communities?

Answering these questions may provide insights into both player behavior and the effectiveness of ranking-based competitive systems.

2.2 Data Understanding

The study uses data collected from two independent competitive communities, referred to as **Community A** and **Community B**.

After preprocessing, the available data consists of:

- **107** ranked Community A players;
- **97** ranked Community B players;
- more than **34,000** historical rating observations.

The available information can be grouped into three main categories:

- **Player Statistics:** cumulative performance indicators such as goals, assists, saves, wins, losses, draws, and matches played.
- **Ranking Information:** current rating values and ranking-related metrics.
- **Historical Rating Data:** temporal records describing rating evolution over time.

The resulting datasets support two complementary forms of analysis:

- **Cross-sectional analysis**, which compares players at a specific point in time.
- **Longitudinal analysis**, which investigates rating progression throughout a player’s competitive activity.

Before proceeding with modeling, descriptive statistics and exploratory visualizations will be used to evaluate data quality, identify potential anomalies, and understand the overall structure of the datasets.

2.3 Data Preparation

Before performing any analysis, the raw data underwent a preprocessing phase designed to improve consistency, reliability, and privacy.

Anonymization

All player identifiers were anonymized by replacing original UUIDs with artificial identifiers (e.g., Player_001, Player_002, etc.). This procedure preserves privacy while maintaining the possibility of tracking player activity across datasets.

Data Cleaning

Inactive players with no ranked activity were removed from the analysis. Administrative fields unrelated to competitive performance were discarded, and missing values were handled according to the semantic meaning of each variable.

Additionally, anomalous rating values were identified and validated through consistency checks against the available rating history.

Data Transformation

Community A dataset originally stored player statistics in a long format, where each row contained a statistic type and its corresponding value. A pivot transformation was therefore applied to obtain a player-level representation suitable for statistical analysis.

Community B dataset was already available in this format and consequently required no equivalent transformation.

Timestamp information was converted into a standard datetime representation, and historical rating records were organized chronologically for each player. A relative temporal variable was also computed, representing the number of days elapsed since each player's first recorded rating observation.

Feature Engineering

To enable fair comparisons between players with different activity levels, several normalized performance indicators were introduced:

- **Win Rate**
- **Goals per Match**
- **Assists per Match**
- **Saves per Match**
- **Passes per Match** (Community B only)

Output Datasets

The preprocessing phase produced four cleaned datasets:

- Community A Player Dataset
- Community A Rating History Dataset
- Community B Player Dataset
- Community B Rating History Dataset

These datasets provide a consistent and reproducible representation of player performance and rating progression, suitable for statistical analysis, dimensionality reduction, clustering, and temporal modeling.

2.4 Modeling

The analytical phase employs a combination of statistical and machine learning techniques.

First, descriptive statistics are used to summarize the main characteristics of the datasets and identify relevant distributional properties.

Correlation analysis is then performed to investigate relationships between performance indicators and player ratings.

To explore the underlying structure of the data, **Principal Component Analysis (PCA)** is applied as a dimensionality reduction technique. Clustering algorithms are subsequently used to identify groups of players with similar statistical profiles.

Finally, temporal analyses are conducted on the rating-history datasets in order to investigate ranking progression and rating stability over time.

2.5 Evaluation

The quality and usefulness of the obtained results are assessed through both quantitative and qualitative criteria.

For correlation analysis, the strength and direction of the observed relationships are evaluated. For dimensionality reduction, the proportion of explained variance is considered. Clustering results are assessed according to their interpretability and their ability to identify meaningful player groups.

The temporal analyses are evaluated according to their ability to describe rating evolution and highlight differences between player trajectories.

The final evaluation focuses on determining whether the identified patterns provide meaningful insights into competitive behavior and ranking-system dynamics.

3 Exploratory Data Analysis

3.1 Dataset Overview

The analysis is based on data collected from two independent competitive multiplayer communities, referred to as **Community A** and **Community B**. Both communities employ rating-based matchmaking systems; however, they use different MMR update formulas and ranking mechanisms. Consequently, direct comparisons of absolute rating values should be interpreted with caution, as differences in rating distributions may be influenced not only by player performance but also by the characteristics of the underlying ranking systems.

After preprocessing and data cleaning, the final datasets contain 107 ranked players from Community A and 97 ranked players from Community B. In addition to player-level statistics, the study includes more than 34,000 historical rating observations collected throughout multiple competitive seasons. These data provide a sufficiently large sample for both statistical and temporal analyses.

The available information can be divided into three main categories:

- **Player Statistics:** cumulative indicators such as matches played, wins, losses, goals, assists, saves, and other performance-related metrics.
- **Rating Information:** current MMR values and ranking-related variables.
- **Historical Rating Records:** chronological observations describing rating progression over time.

Table 1 summarizes the main characteristics of the analyzed datasets.

Table 1: Overview of the analyzed datasets.

Metric	Community A	Community B
Players	107	97
Ranking system	MMR Formula A	MMR Formula B
Analysis type	Cross-sectional and temporal	Cross-sectional and temporal

The datasets support two complementary forms of analysis. Cross-sectional analyses compare players at a given point in time using aggregated performance statistics, whereas longitudinal analyses investigate how player ratings evolve throughout their competitive activity. The combination of these perspectives enables the study of both player behavior and ranking-system dynamics.

Before applying statistical and machine learning techniques, an exploratory analysis was performed to understand the overall structure of the datasets, identify potential anomalies, evaluate data quality, and characterize the populations under investigation. The following sections examine rating distributions and player activity patterns, providing an initial understanding of the competitive ecosystems represented by the two communities.

3.2 Rating Distribution

The first exploratory step consists of analyzing the distribution of player ratings (MMR) within the two communities.

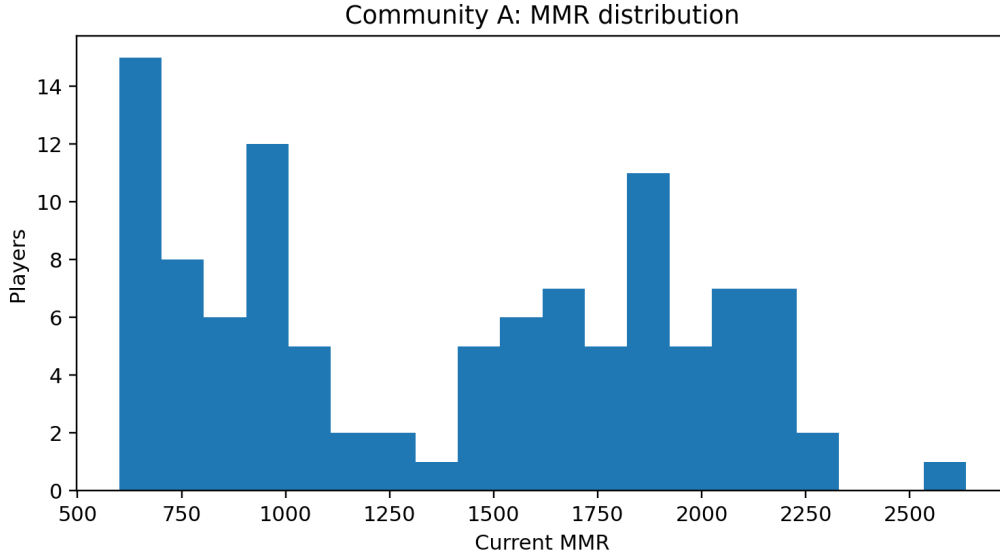


Figure 1: MMR distribution in Community A.

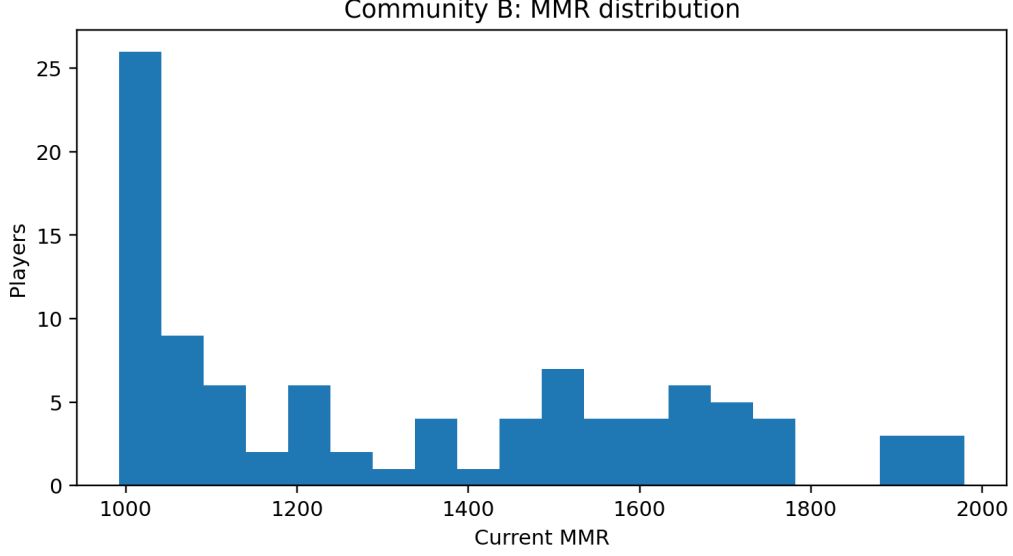


Figure 2: MMR distribution in Community B.

The distributions reveal substantial variability in player ratings across both communities.

Community A exhibits a particularly wide rating range, extending from approximately 600 to more than 2500 MMR. The distribution appears highly heterogeneous and suggests the presence of distinct competitive strata within the player population. A large concentration of players is observed at lower rating levels, while a second concentration emerges among highly rated competitors. This pattern may indicate a separation between casual participants and highly competitive players.

Community B also presents a non-uniform rating distribution but appears considerably more compact. Most players are concentrated within a narrower rating interval, resulting in lower overall rating dispersion. Although variability remains present, the degree of stratification appears less pronounced than in Community A.

Neither distribution closely resembles a Gaussian distribution. The observed asymmetries suggest that rating dynamics are influenced by multiple factors, including player activity, matchmaking behavior, community structure, and the specific characteristics of the ranking algorithms.

It is important to emphasize that the two communities employ different MMR formulas. Consequently, differences in rating ranges and distribution shapes cannot be interpreted exclusively as differences in player skill. Part of the observed variability may originate from the mathematical properties of the rating systems themselves.

Overall, the rating distributions suggest that both communities contain players with substantially different competitive levels, although Community A appears to exhibit a stronger degree of rating stratification and rating dispersion.

3.3 Player Activity Distribution

Player activity was analyzed through the total number of ranked matches played by each participant.

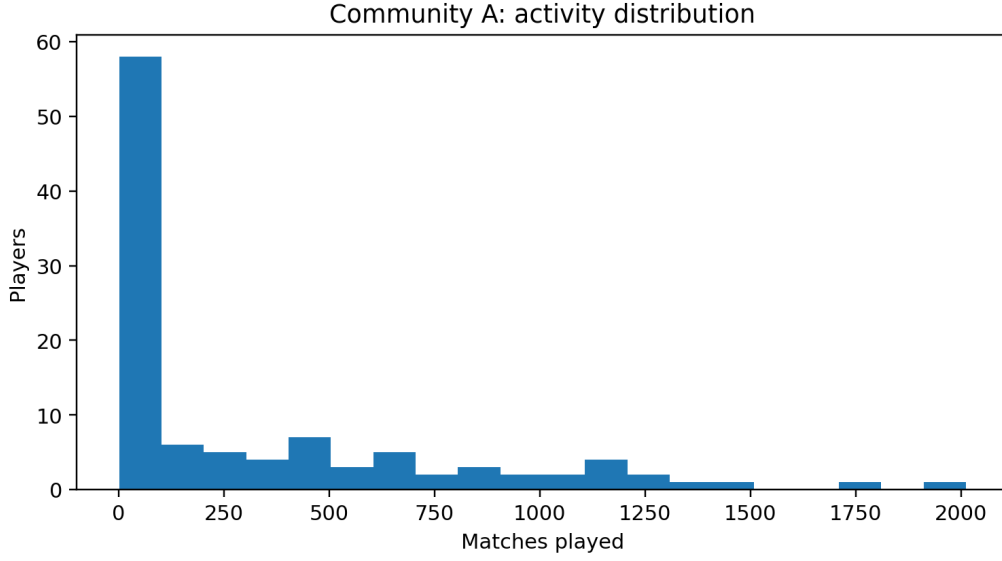


Figure 3: Distribution of matches played in Community A.

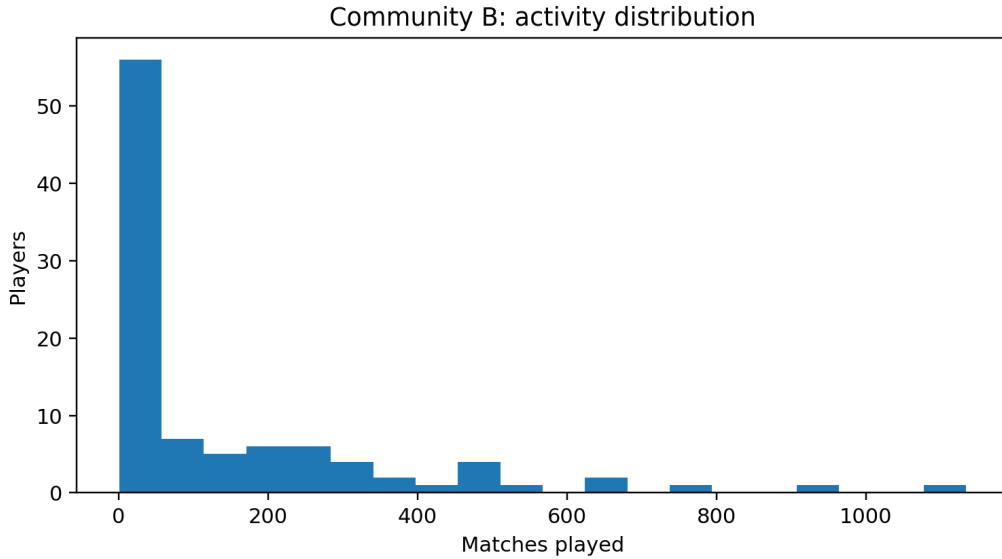


Figure 4: Distribution of matches played in Community B.

Both communities display a strongly right-skewed distribution of player activity.

The majority of players participate in a relatively small number of ranked matches, while a limited number of highly active players account for a disproportionately large amount of competitive activity. This phenomenon is commonly referred to as a *long-tail distribution* and is frequently observed in online gaming ecosystems.

Community A exhibits a particularly pronounced long tail, with several players exceeding one thousand ranked matches and some approaching two thousand matches. This indicates the presence of a small but highly engaged group of veteran players who contribute substantially to the competitive environment and generate a large fraction of the available data.

Community B shows a similar pattern, although the tail is less extreme. While highly active players are still present, the overall activity distribution appears more concentrated

and less polarized than in Community A.

The existence of a small number of extremely active participants alongside a large population of casual players suggests that activity is unevenly distributed across both communities. Such imbalances are common in competitive online systems and may significantly affect ranking progression, player development, and competitive outcomes.

For this reason, subsequent analyses will investigate whether highly ranked players achieve their position primarily through superior performance, increased participation, or a combination of both factors.

3.4 Community Comparison

To provide a direct comparison between the two competitive environments, the main descriptive statistics of the analyzed variables are reported in Table 2.

Table 2: Comparison of key statistics between the two communities.

Metric	Community A	Community B
Average MMR	1387.6	1329.6
MMR Std. Dev.	560.2	302.7
Average Matches	323.4	140.9
Median Matches	65	27
Win Rate (%)	49.1	52.8
Goals per Match	0.432	0.486
Assists per Match	0.312	0.300
Saves per Match	0.846	0.684

Several differences emerge from the comparison.

First, Community A appears substantially more active than Community B. Players in Community A participate in an average of 323 ranked matches, compared to approximately 141 matches in Community B. The difference is also visible in the median values (65 versus 27 matches), suggesting that higher activity levels are not limited to a few extreme players but characterize the community as a whole.

Second, Community A exhibits a considerably wider rating distribution. Although the average MMR values are relatively similar, the standard deviation of MMR in Community A is almost twice that of Community B (560 versus 303). This result is consistent with the rating-distribution analysis presented previously and suggests a stronger degree of competitive stratification.

Performance indicators show smaller differences between the two communities. Community B exhibits a slightly higher average win rate (52.8% versus 49.1%) and a marginally higher number of goals per match. Conversely, Community A records a higher average number of saves per match, possibly reflecting differences in gameplay dynamics or player roles.

It is important to emphasize that the two communities employ different MMR formulas and ranking systems. Consequently, direct comparisons of absolute rating values should be interpreted carefully. Some observed differences may originate from the mathematical properties of the ranking algorithms rather than from differences in player skill alone.

Overall, the comparison suggests that Community A is characterized by higher player engagement and greater rating dispersion, whereas Community B appears more compact

and homogeneous.

4 Statistical Analysis

4.1 Correlation Analysis

To investigate the relationships between player performance indicators and ranking outcomes, Pearson correlation matrices were computed for both communities. The analysis was performed on the complete datasets and repeated after progressively filtering out less active players (20+, 30+, 40+, and 50+ matches).

Figure 5 shows the correlation matrices obtained from the complete datasets.

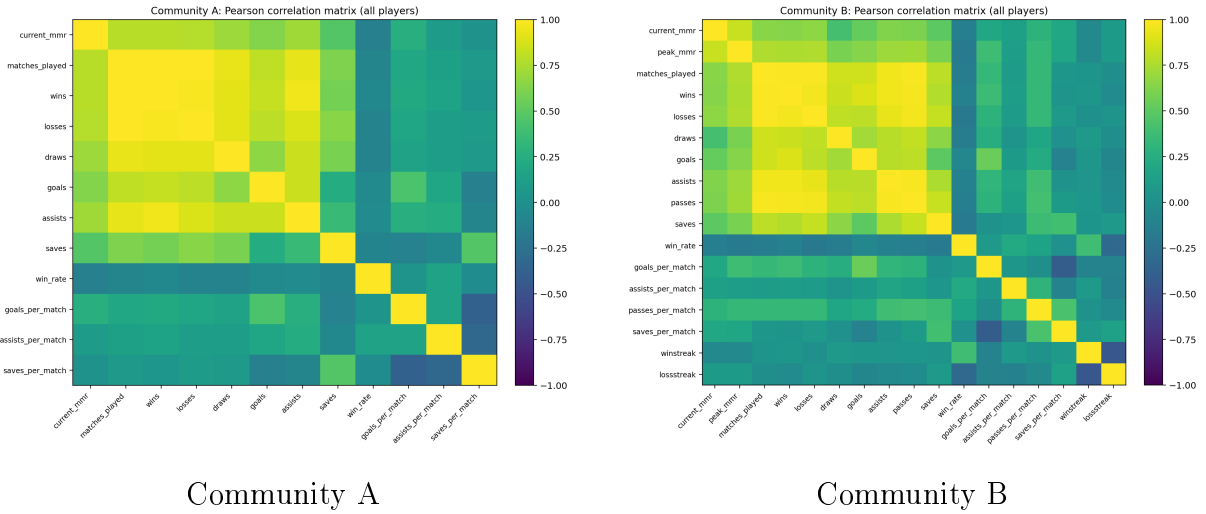


Figure 5: Pearson correlation matrices computed on the complete datasets.

Several common patterns emerge across the two communities. Current rating is positively associated with offensive statistics such as goals and assists, while activity-related variables (matches played, wins, and total contributions) are strongly correlated with one another. This is expected because highly active players naturally accumulate larger absolute statistics over time.

A notable result is the relatively weak relationship between win rate and current rating. Despite being an intuitive indicator of performance, win rate alone does not appear to explain ranking differences particularly well. This can be attributed to the rating systems used by the two communities, which reward long-term performance and consistency rather than short-term success.

The strongest relationships with rating are generally found in offensive metrics and overall participation. Goals, assists, and total activity consistently display positive associations with ranking outcomes, suggesting that sustained contribution over many matches is a key factor in reaching higher competitive tiers.

To evaluate the robustness of these relationships, the analysis was repeated using only increasingly active players. Figures 6 and 7 summarize the resulting correlation matrices.

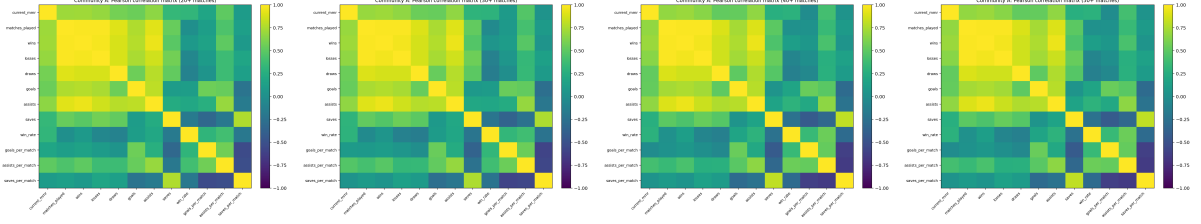


Figure 6: Community A correlation matrices considering increasingly active players (20+, 30+, 40+, and 50+ matches).

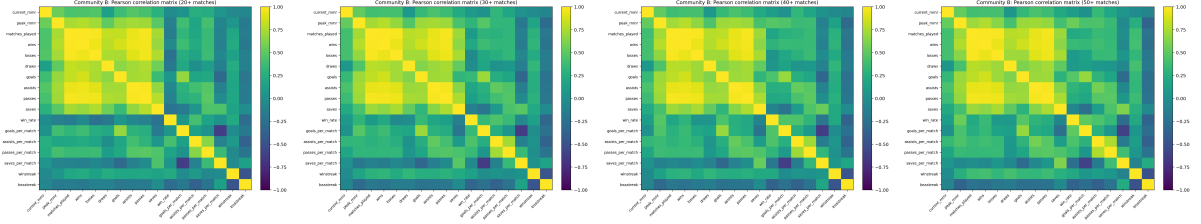


Figure 7: Community B correlation matrices considering increasingly active players (20+, 30+, 40+, and 50+ matches).

The threshold analysis reveals that the overall correlation structure remains remarkably stable as low-activity players are removed. This suggests that the observed relationships are not driven by occasional participants or statistical noise. Instead, they appear to reflect persistent behavioral patterns within the competitive ecosystems.

An interesting observation concerns defensive statistics. While total saves show a moderate positive association with rating, saves per match tend to exhibit weak or negative relationships with ranking outcomes, particularly among highly active players. A possible interpretation is that stronger teams spend less time defending and therefore generate fewer saving opportunities per game. Consequently, high save rates may reflect defensive pressure rather than superior overall performance.

Overall, the correlation analysis indicates that ranking is influenced more strongly by long-term participation and offensive contribution than by isolated efficiency metrics. The consistency of the results across different activity thresholds further supports the reliability of these findings.

4.2 Activity versus Skill

A central question in ranking analysis is whether highly rated players are actually more skilled or simply more active. To investigate this aspect, the relationship between the number of matches played, current rating and win rate was analyzed for both communities.

Table 3: Pearson correlations between player activity, current rating and win rate.

Dataset	Relationship	Pearson correlation
Community A	Matches played vs current MMR	0.784
Community A	Matches played vs win rate	-0.089
Community B	Matches played vs current MMR	0.645
Community B	Matches played vs win rate	-0.162

Table 3 shows that player activity is strongly associated with current rating. The correlation between matches played and current MMR is high in Community A and moderately high in Community B. This indicates that more active players tend to reach higher ranking positions.

However, the relationship between activity and win rate is close to zero and slightly negative in both communities. This suggests that playing more matches does not necessarily correspond to maintaining a higher winning percentage. This behaviour is consistent with Elo-like ranking systems: as players improve and climb the ranking ladder, they are progressively matched against stronger opponents, causing their win rate to stabilize rather than increase indefinitely.

Figure 8 shows the scatter plots between matches played and current rating.

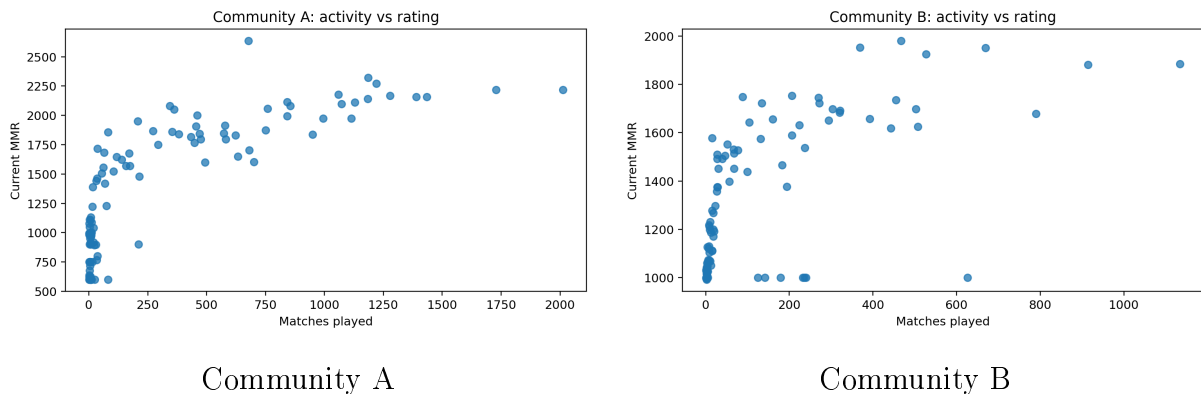


Figure 8: Relationship between player activity and current rating in the two communities. Each point represents one player.

The scatter plots confirm the correlation results. In both communities, players with very low activity are concentrated around lower or intermediate ratings, while the highest ratings are generally reached by players with a larger number of matches. At the same time, the relationship is not deterministic: some active players remain at relatively low ratings, while some less active players achieve competitive ratings.

This suggests that activity is an important condition for ranking progression, but it is not sufficient by itself. High activity provides more opportunities to gain rating, but competitive success still depends on performance, matchmaking dynamics and consistency over time.

Overall, the analysis indicates that rating is strongly linked to long-term participation, whereas win rate alone does not fully capture player skill. In this context, current MMR should be interpreted as a combined indicator of experience, consistency and competitive success rather than as a direct measure of raw winning percentage.

4.3 High-Ranked versus Low-Ranked Players

To better understand which characteristics distinguish highly ranked players from the rest of the population, players were divided into three groups according to their current MMR: the bottom 25%, the middle 50%, and the top 25%. This grouping allows a direct comparison between low-ranked, average-ranked and high-ranked players.

Table 4 reports the average values of the main normalized performance indicators for each rank group.

Table 4: Average performance indicators by rank group.

Dataset	Rank group	Win rate	Goals/match	Assists/match	Saves/match
Community A	Bottom 25%	0.56	0.31	0.30	0.89
Community A	Middle 50%	0.46	0.42	0.28	0.81
Community A	Top 25%	0.48	0.59	0.39	0.86
Community B	Bottom 25%	0.53	0.53	0.30	0.32
Community B	Middle 50%	0.52	0.36	0.28	0.77
Community B	Top 25%	0.46	0.69	0.35	0.89

The results show that high-ranked players are not simply characterized by higher win rates. In both communities, win rate varies only moderately across rank groups and does not display a clear monotonic increase. This suggests that current rating is not merely a direct transformation of the fraction of matches won.

A clearer separation emerges from performance indicators normalized by match count. In Community A, the top 25% of players show higher goals per match and assists per match than the lower-ranked groups. This indicates that high-ranked players tend to contribute more consistently to offensive actions, both by scoring and by supporting teammates.

In Community B, the distinction is even broader. Top-ranked players exhibit higher goals per match and assists per match, but they also show higher saves per match. This suggests that in Community B, high-ranked players may contribute more across multiple phases of the game, including both offensive and defensive actions.

Figure 9 shows the distributions of normalized performance metrics across rank groups for Community A.

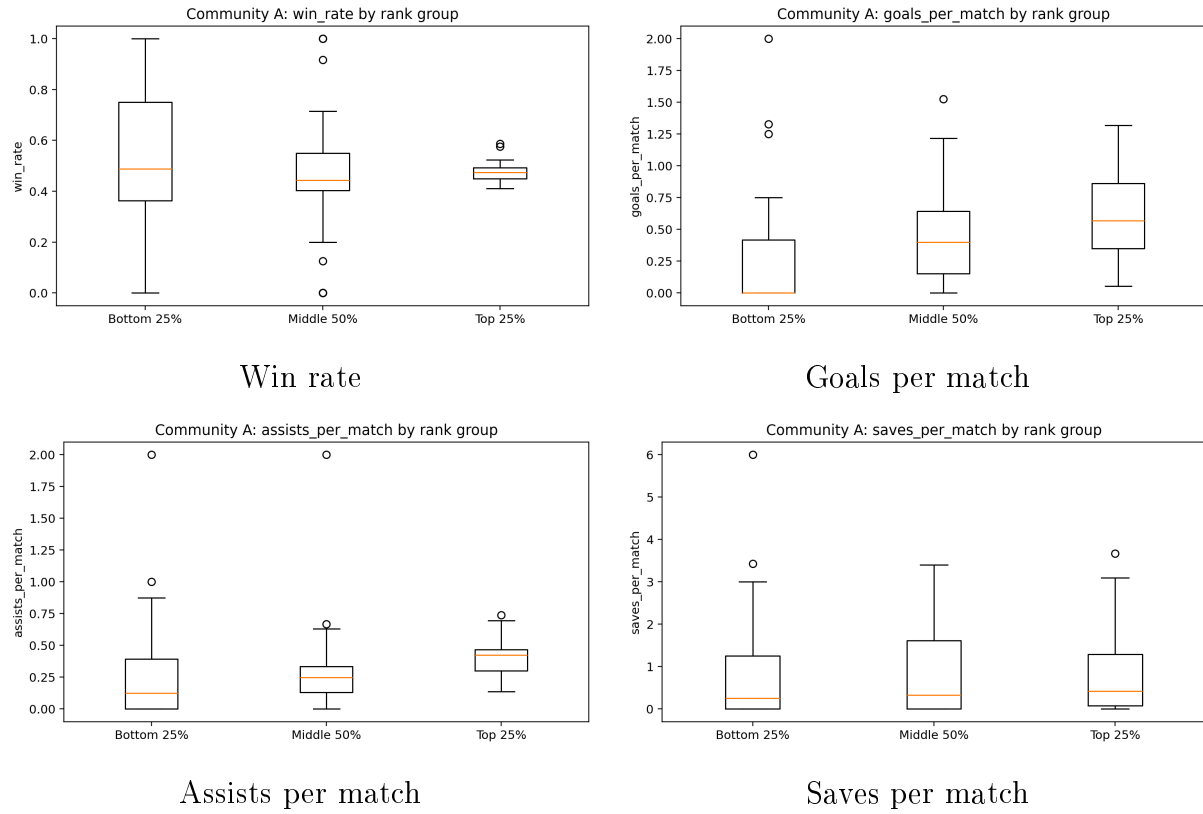


Figure 9: Distribution of normalized performance indicators by rank group in Community A.

Figure 10 reports the corresponding analysis for Community B.

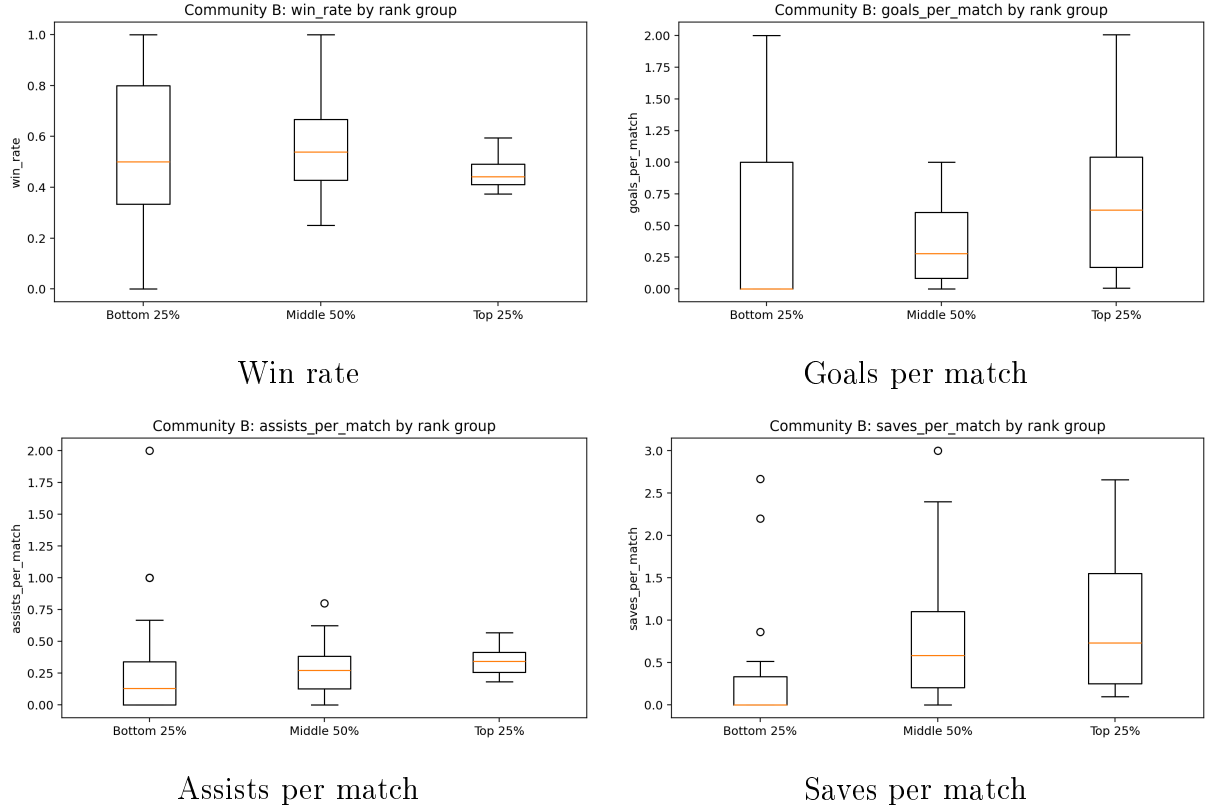


Figure 10: Distribution of normalized performance indicators by rank group in Community B.

The boxplots confirm that offensive contribution is one of the clearest differences between rank groups. Goals per match increase from lower-ranked to higher-ranked players, especially in Community A. Assists per match also tend to be higher among top-ranked players, suggesting that high-level performance is associated not only with individual scoring but also with team-oriented contribution.

The behaviour of saves per match differs between the two communities. In Community A, saves per match are relatively similar across groups, suggesting that defensive actions do not strongly distinguish rank levels. In Community B, instead, saves per match increase noticeably among higher-ranked players, indicating that defensive contribution may be more relevant in that environment.

Overall, the analysis suggests that high-ranked players are better characterized by consistent in-game contribution than by win rate alone. Offensive production, especially goals and assists per match, appears to be the most stable discriminator between low-ranked and high-ranked players, while defensive contribution depends more strongly on the specific community and gameplay structure.

5 Temporal Analysis

Temporal analysis was performed to investigate how player ratings evolved throughout the observation period and to evaluate the stability of the ranking systems adopted by the two communities. While previous sections focused on static properties of player performance, this section examines the dynamic behavior of ratings over time.

5.1 Rating Evolution Over Time

To study the long-term behavior of the ranking systems, the average rating of all active players was computed for each day in the observation period. Figures 11 and 12 show the resulting aggregate rating trajectories.

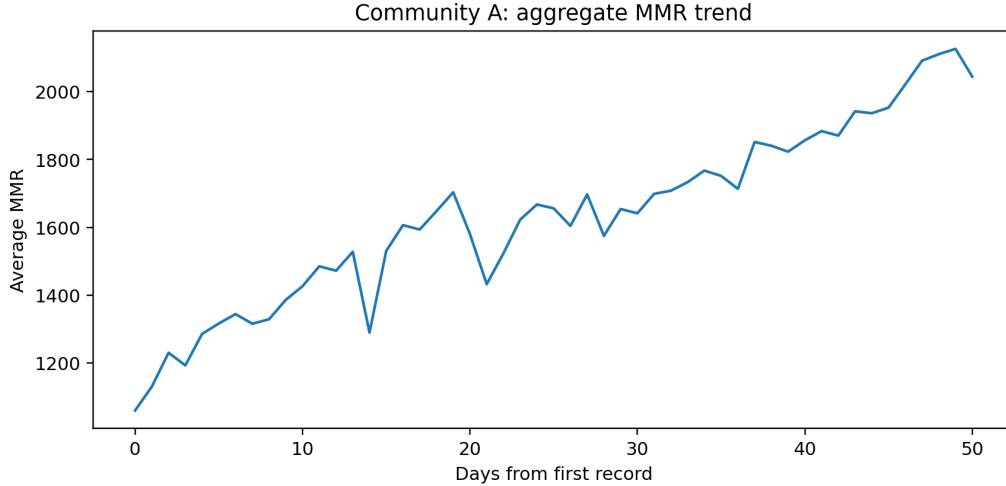


Figure 11: Aggregate rating trend in Community A.

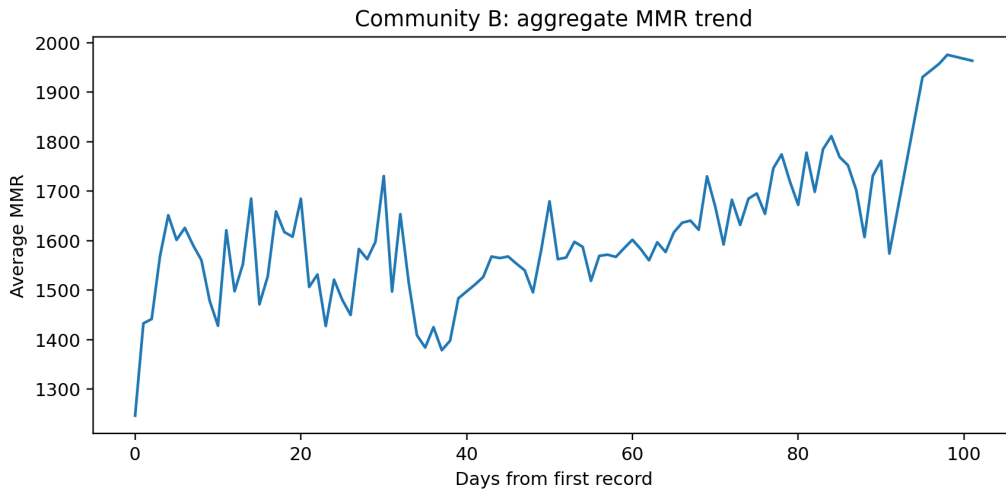


Figure 12: Aggregate rating trend in Community B.

Both communities exhibit a clear upward trend in average rating over time. Community A shows a particularly strong increase, with the average rating rising from approximately 1050 MMR to over 2000 MMR during the observation period. Community B displays a similar pattern, although the growth is less pronounced and characterized by greater short-term fluctuations.

The presence of a persistent increase in average rating suggests the existence of rating inflation, meaning that the total amount of rating distributed within the system tends to grow over time rather than remaining centered around a stable equilibrium. This phenomenon appears to be stronger in Community A than in Community B.

Overall, the temporal evolution indicates that both ranking systems successfully differentiate players over time, but they do not maintain a perfectly stationary rating distribution.

5.2 Rating Stability

To evaluate rating stability, the standard deviation of each player’s rating history was computed. Higher values indicate stronger fluctuations over time, whereas lower values correspond to more stable ratings.

Table 5: Summary of rating volatility in the two communities.

Metric	Community A	Community B
Players analysed	107	97
Observation period (days)	51	101
Overall volatility level	Higher	Lower
Rating fluctuations	Larger	More moderate
System behaviour	More dynamic	More stable

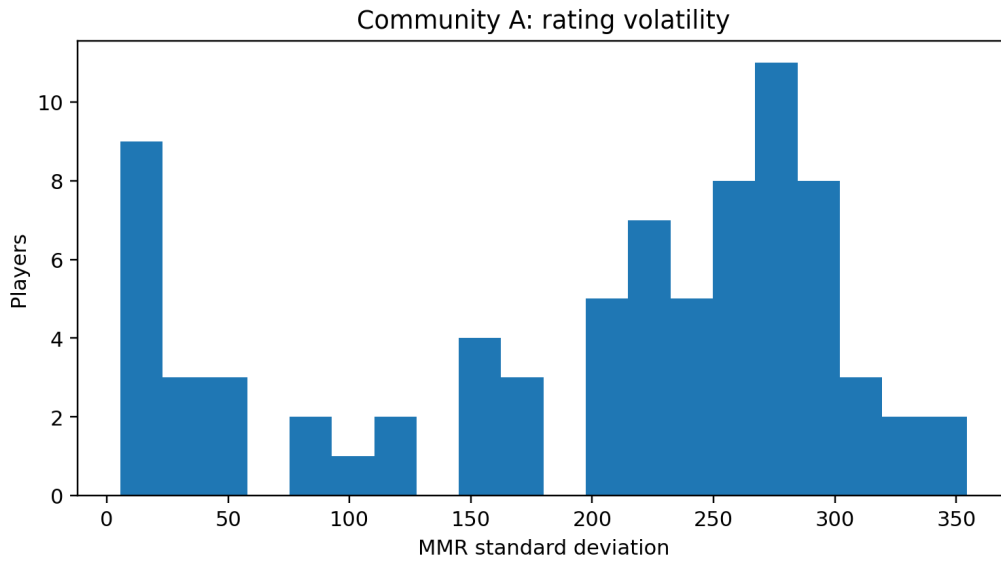


Figure 13: Distribution of rating volatility in Community A.

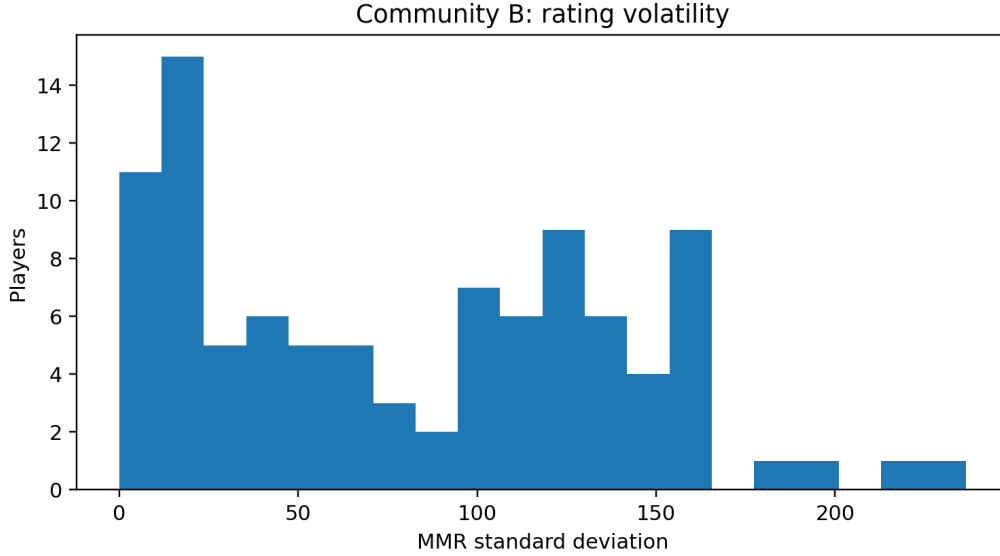


Figure 14: Distribution of rating volatility in Community B.

The volatility distributions reveal meaningful differences between the two communities. Community A exhibits a broader spread of rating fluctuations, with many players experiencing large variations throughout their competitive history. This suggests a ranking environment where ratings can change rapidly in response to performance.

Community B displays a more concentrated volatility distribution. Most players remain within a narrower range of rating variation, indicating that ratings evolve more gradually over time. Such behaviour is generally associated with a more stable ranking system.

Although both communities show evidence of rating inflation and long-term growth, Community B appears to maintain greater rating stability. Community A, on the other hand, produces larger rating movements and therefore a more dynamic competitive environment.

It is important to note that the two communities employ different rating formulas. Consequently, these observations should not be interpreted as proof that one system is objectively superior, but rather as evidence that the systems emphasize different trade-offs between responsiveness and stability.

6 Advanced Analytics

6.1 Principal Component Analysis

Principal Component Analysis (PCA) was applied to reduce the dimensionality of the player statistics and to identify the main latent dimensions behind player behaviour. The variable *current MMR* was excluded from the PCA input, so that the components describe gameplay profiles rather than being directly dominated by the rating itself.

Dataset	PC1	PC2
Community A	32.64%	22.67%
Community B	27.64%	25.75%

Table 6: Explained variance ratio of the first two principal components.

The first two components explain approximately 55.31% of the total variance in Community A and 53.39% in Community B. This indicates that the two-dimensional projections preserve a substantial part of the information contained in the original player statistics, while still simplifying the interpretation of player profiles.

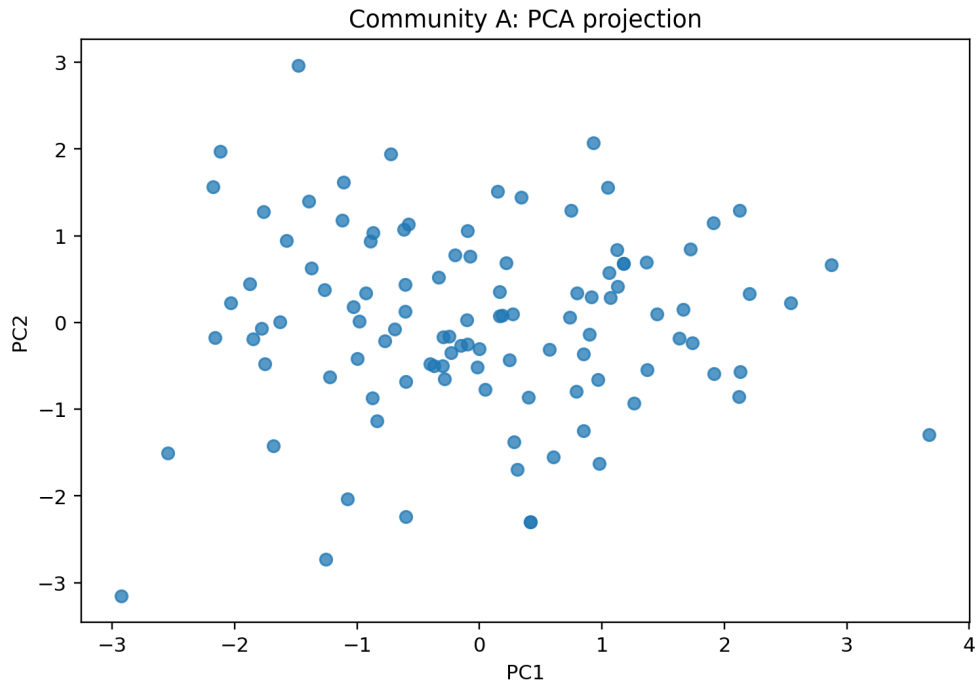


Figure 15: PCA projection of players in Community A.

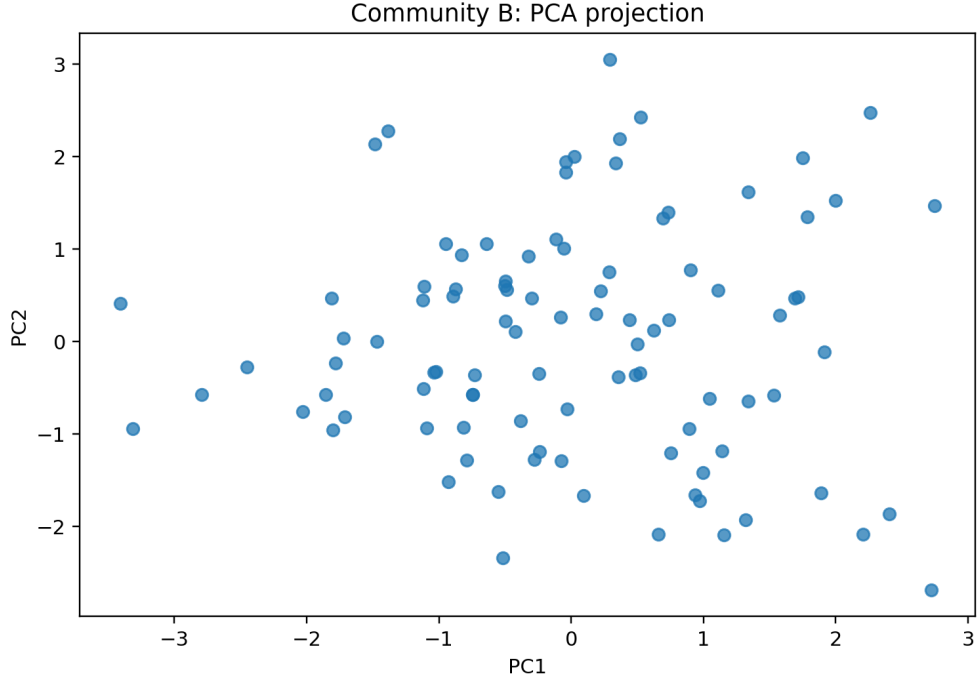


Figure 16: PCA projection of players in Community B.

To interpret the axes, the component loadings were inspected. In Community A, PC1 mainly separates offensive and defensive contributions: goals and assists have negative loadings, while saves have a positive loading. PC2 is mainly associated with activity and experience, with a strong positive loading for matches played and a negative loading for win rate.

Feature	PC1	PC2
Matches played	-0.201	0.717
Win rate	-0.158	-0.614
Goals per match	-0.555	0.238
Assists per match	-0.527	-0.135
Saves per match	0.591	0.183

Table 7: PCA loadings for Community A.

In Community B, PC1 is mostly related to overall in-game involvement, especially passes, saves, assists, and matches played. PC2 instead separates offensive and defensive contribution, with goals loading positively and saves loading negatively.

Feature	PC1	PC2
Matches played	0.369	0.397
Win rate	0.208	0.084
Goals per match	-0.046	0.687
Assists per match	0.368	0.286
Saves per match	0.450	-0.531
Passes per match	0.693	0.003

Table 8: PCA loadings for Community B.

Overall, both communities show similar latent dimensions: one related to activity or general involvement, and another related to the type of in-game contribution, especially

the distinction between offensive and defensive profiles. The PCA projections do not show clearly separated groups of players; instead, players are distributed along continuous dimensions. This suggests that player behaviour varies gradually rather than forming naturally distinct categories.

6.2 Player Clustering

To identify recurring player archetypes, K-Means clustering was applied to the standardized player statistics. The number of clusters was fixed at $k = 4$ for both communities. Clustering was performed independently for each dataset using the same gameplay-related variables considered in the PCA analysis.

The resulting clusters were projected into the two-dimensional PCA space to facilitate visualization and interpretation.

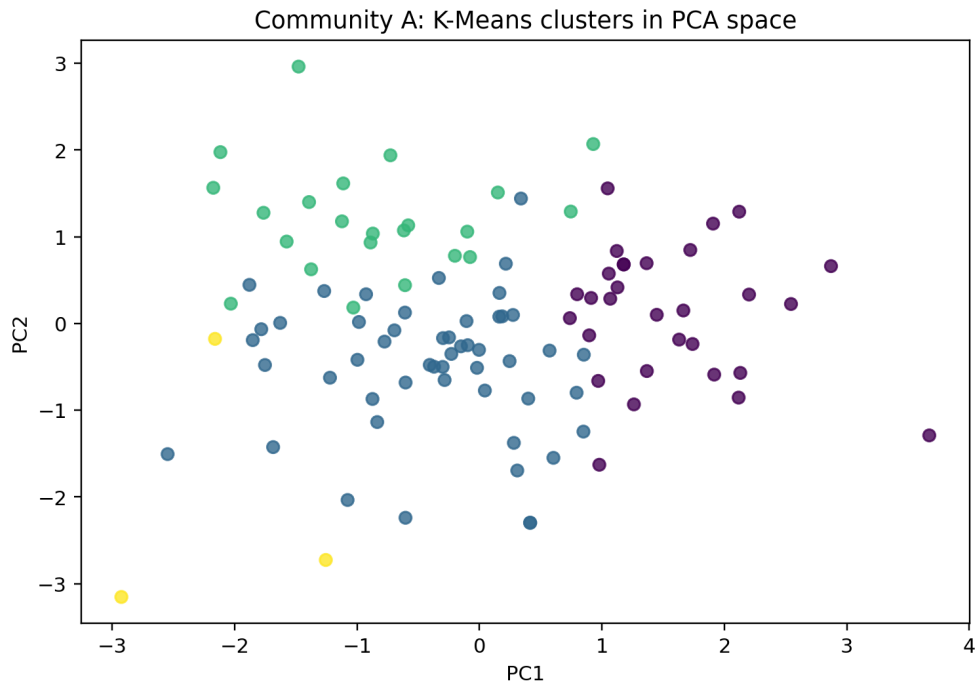


Figure 17: K-Means clusters projected in PCA space for Community A.

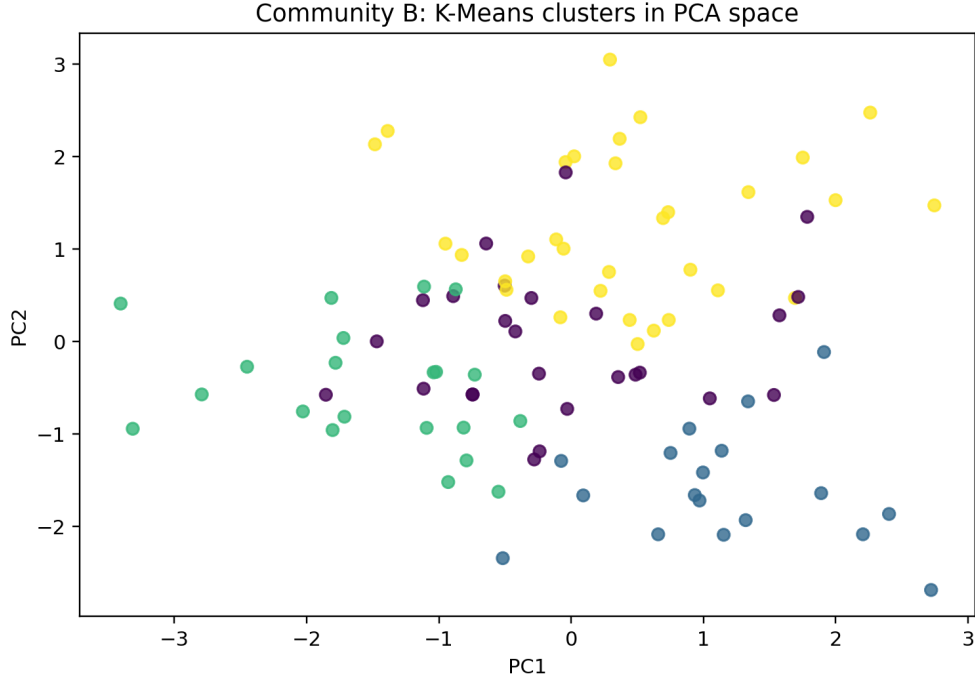


Figure 18: K-Means clusters projected in PCA space for Community B.

To evaluate clustering quality, the silhouette coefficient was computed for each dataset.

Dataset	Silhouette Score
Community A	0.244
Community B	0.230

Table 9: Clustering quality measured using the silhouette coefficient.

The silhouette scores indicate moderate separation between clusters rather than strongly isolated groups. Nevertheless, the resulting clusters remain highly interpretable and provide useful insight into recurring player profiles.

For Community A, the clustering process identified four distinct archetypes. The first group consists of defensive specialists characterized by a very high number of saves per match and relatively low offensive statistics. A second group is composed of offensive players, who achieve higher numbers of goals and assists while contributing less defensively. A third cluster contains highly experienced competitors with exceptionally high MMR values and extensive match histories, representing the elite veteran players of the community. Finally, a small cluster is formed by new or inactive players, whose statistics are based on very limited match participation.

For Community B, four comparable profiles emerged. One cluster is clearly associated with goalkeepers and defensive players, exhibiting the highest save rates in the dataset. Another cluster contains highly active attacking players with the highest goal production and the highest average MMR. A third cluster consists of casual or lower-performing players, characterized by lower activity and lower win rates. The final cluster contains emerging players who have relatively few matches played but unusually strong win rates, suggesting promising early performance that has not yet stabilized.

The clustering results are consistent with the PCA findings. In both communities, PCA revealed two dominant latent dimensions: overall activity or involvement and offensive-versus-defensive contribution. The clusters largely follow these same dimensions, separating goalkeepers from attacking players and highly active competitors from low-activity participants.

Overall, the clustering analysis suggests that player behaviour is not randomly distributed but organized around a small number of recurring statistical archetypes. Despite the differences between the two communities, similar profiles emerge in both datasets, indicating common patterns of participation, specialization, and competitive development.

6.3 Outlier Detection

Outlier detection was performed to identify players whose ranking statistics deviate from the general relationship between current MMR and win rate. For each player, both variables were standardized using z-scores, and an outlier score was computed as the absolute difference between standardized MMR and standardized win rate. A high score therefore indicates a strong mismatch between a player’s rating and their observed win rate.

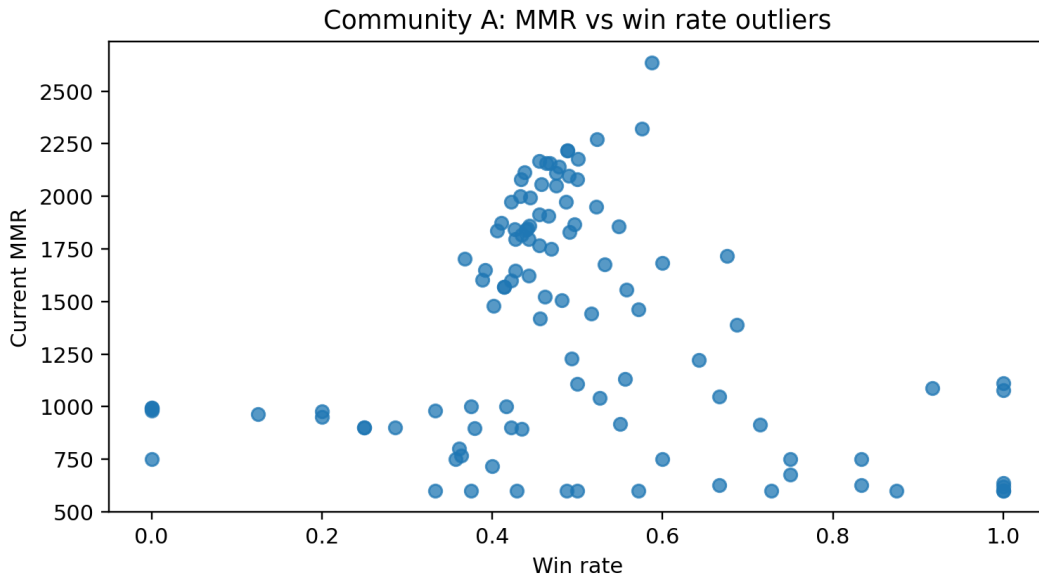


Figure 19: Outlier detection based on current MMR and win rate in Community A.

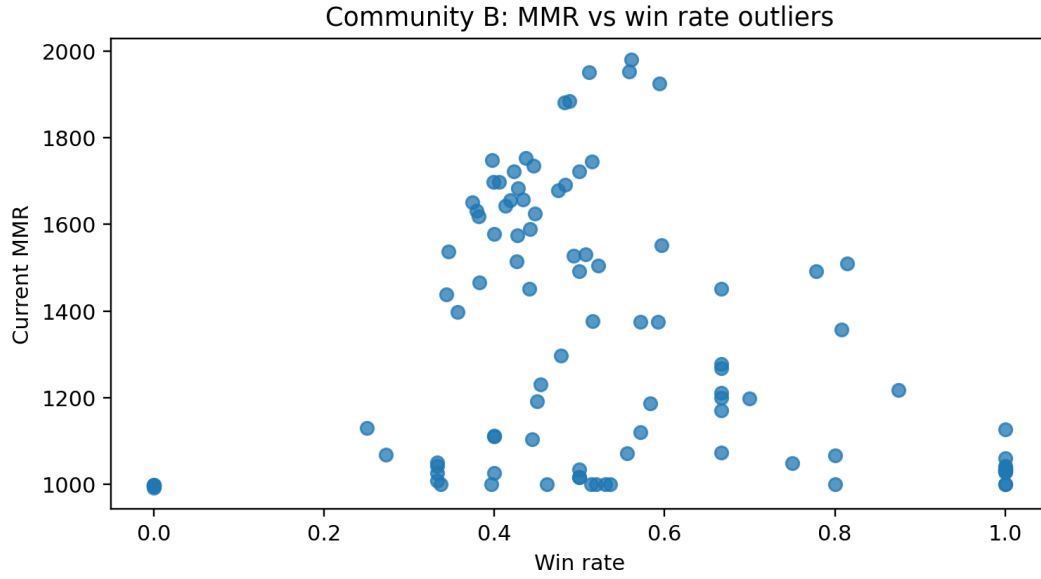


Figure 20: Outlier detection based on current MMR and win rate in Community B.

The scatter plots show that most players follow a broad positive relationship between win rate and current MMR: players with higher win rates generally tend to have higher ratings. However, the relationship is not perfectly linear, especially for players with very high or very low win rates.

The detected outliers are mostly players whose win rate is extreme despite having relatively low or initial MMR values. This pattern is particularly visible for players with a win rate close to 100% but an MMR near the lower part of the distribution. Rather than indicating abnormal competitive behaviour, these cases are likely explained by low activity: players with only a few matches can easily obtain extreme win rates that are not yet statistically stable.

Therefore, the outlier analysis highlights an important limitation of ranking data: win rate can be misleading when the number of matches played is small. In this sense, the analysis did not reveal strong evidence of problematic or suspicious players, but it showed that sample size must be considered when interpreting player performance.

Dataset	Player	Current MMR	Win Rate	Matches Played	Outlier Score
Community A	Player_084	600.0000	1.0000	1	3.8819
Community A	Player_152	600.0000	1.0000	1	3.8819
Community A	Player_010	620.0000	1.0000	2	3.8460
Community A	Player_006	635.0000	1.0000	1	3.8191
Community A	Player_009	600.0000	0.8750	8	3.2758
Community A	Player_155	625.0000	0.8333	6	3.0289
Community A	Player_018	1077.0000	1.0000	2	3.0264
Community A	Player_037	1110.0000	1.0000	3	2.9672
Community A	Player_093	750.0000	0.8333	6	2.8047
Community A	Player_104	1088.0000	0.9167	12	2.6025
Community B	BB_Player_185	1000.0000	1.0000	1	3.1444
Community B	BB_Player_195	1000.0000	1.0000	1	3.1444
Community B	BB_Player_150	1028.0000	1.0000	1	3.0514
Community B	BB_Player_080	1032.0000	1.0000	1	3.0381
Community B	BB_Player_116	1033.0000	1.0000	1	3.0348
Community B	BB_Player_061	1034.0000	1.0000	2	3.0315
Community B	BB_Player_115	1038.0000	1.0000	2	3.0182
Community B	BB_Player_129	1042.0000	1.0000	2	3.0049
Community B	BB_Player_130	1061.0000	1.0000	3	2.9418
Community B	BB_Player_182	1127.0000	1.0000	4	2.7226

7 Conclusions

This project investigated the behavior of two independent competitive multiplayer ranking systems through a combination of descriptive statistics, correlation analysis, temporal modeling, dimensionality reduction, clustering, and outlier detection.

Several consistent patterns emerged across both communities.

First, player activity was found to be one of the strongest factors associated with ranking progression. Players with higher ratings generally participated in substantially more matches than lower-ranked competitors. However, activity alone was not sufficient to explain ranking outcomes, as the relationship between matches played and win rate was weak in both communities. This suggests that ranking progression is influenced by long-term participation and consistency rather than by short-term success alone.

Second, offensive contribution proved to be one of the most reliable indicators of competitive success. Goals per match and assists per match consistently distinguished highly ranked players from lower-ranked participants, while win rate alone showed limited explanatory power. This indicates that player performance is better captured through sustained in-game contribution than through simple match outcomes.

The dimensionality reduction analysis revealed remarkably similar latent structures in both communities. Despite differences in player populations and ranking formulas, the dominant dimensions were related to overall activity and involvement, together with the distinction between offensive and defensive contribution. These findings suggest that competitive player behavior can be described through a small number of recurring performance dimensions.

The clustering analysis further supported this interpretation. Meaningful player archetypes emerged in both datasets, including offensive players, defensive specialists, highly expe-

rienced veterans, and low-activity participants. The consistency between PCA and clustering results increases confidence that these profiles reflect genuine patterns within the data rather than artifacts of the applied methods.

The temporal analysis revealed clear evidence of rating inflation in both communities, particularly in Community A, where average ratings increased substantially throughout the observation period. At the same time, the two systems exhibited different stability characteristics: Community A appeared more dynamic and volatile, while Community B maintained a more stable rating environment. These results suggest a trade-off between responsiveness and long-term rating stability.

Finally, the outlier analysis showed that most detected anomalies were associated with players having very limited match histories. This highlights the importance of sample size when interpreting ranking statistics and demonstrates that extreme win rates obtained from a small number of matches should be treated with caution.

Overall, the results indicate that the analyzed ranking systems successfully differentiate players according to long-term contribution, participation, and competitive performance. While the presence of rating inflation suggests that improvements could be made to enhance long-term rating stability, the observed statistical patterns are generally consistent with a ranking system that rewards meaningful player contribution rather than relying exclusively on win-loss outcomes.

Despite the differences between the two communities, remarkably similar behavioral structures emerged across both datasets. This suggests that many aspects of competitive player behavior are robust across different multiplayer environments and ranking systems, making them suitable targets for future research in game analytics and competitive matchmaking design.